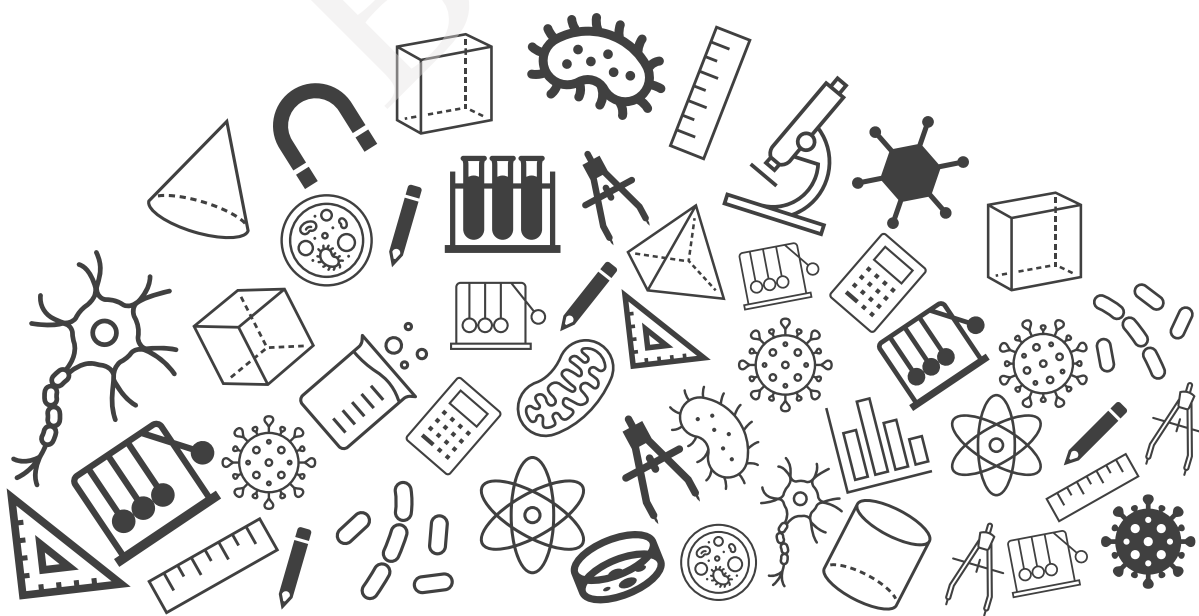




Grade 09

Science Chapter Notes





BYJU'S Classes

Chapter Notes



Force and Laws of Motion

Class IX

Topics to Be Covered

1

1. All about Force

- 1.1 Definition of Force
- 1.2 Effects of force
- 1.3 Types of forces
- 1.4 Balanced and unbalanced force

2

2. Newton's Laws of Motion

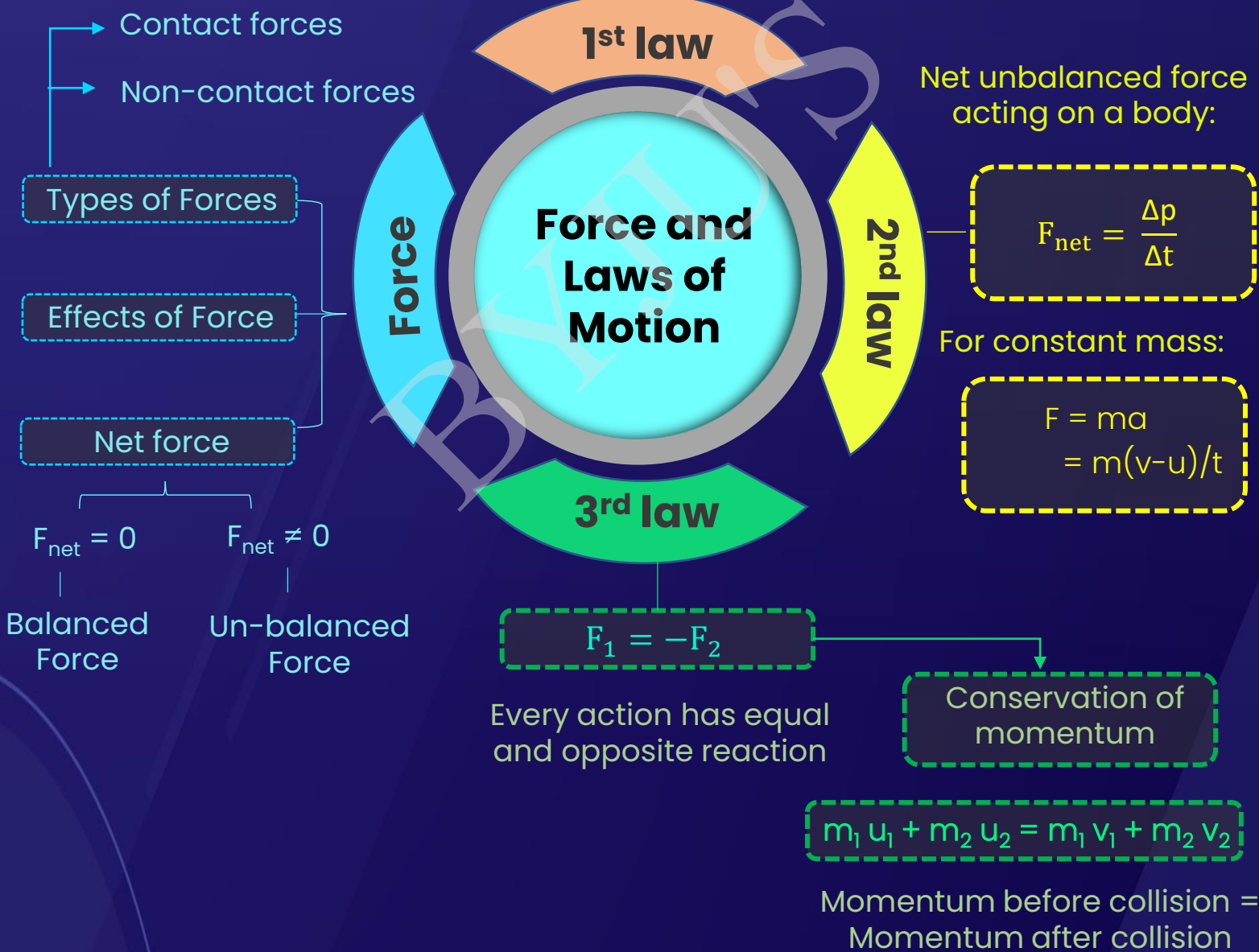
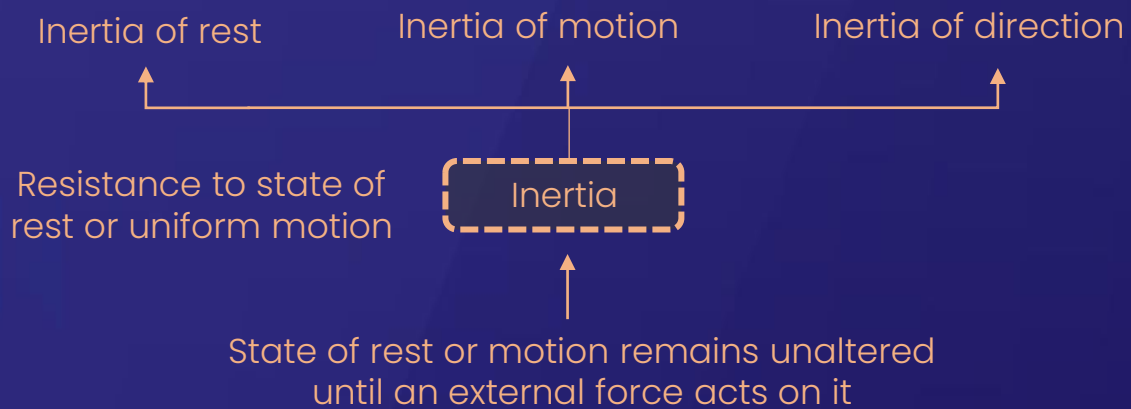
- 2.1 First Law of Motion
- 2.2 Inertia and Mass
- 2.3 Types of Inertia
- 2.4 Momentum
- 2.5 Second Law of Motion
- 2.6 Impulse

2.7 Third law of Motion

- 2.8 Internal and External Forces
- 2.9 Conservation of Momentum



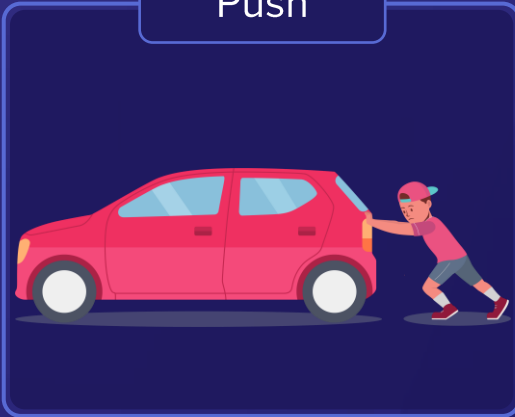
MIND MAP



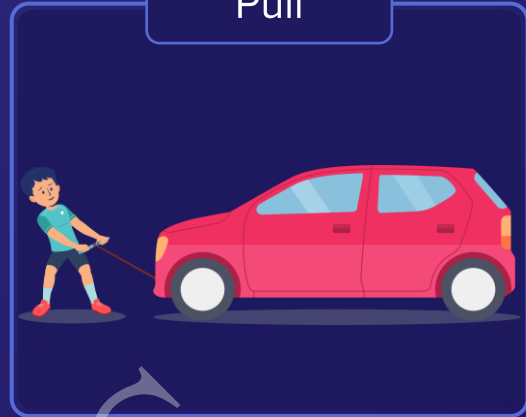
1. Force

1.1 Definition of Force

Push



Pull



- A **Push** or a **Pull** acting on an object is called a Force.
- SI unit of force is **newton(N)**.

1.2 Effects of Force

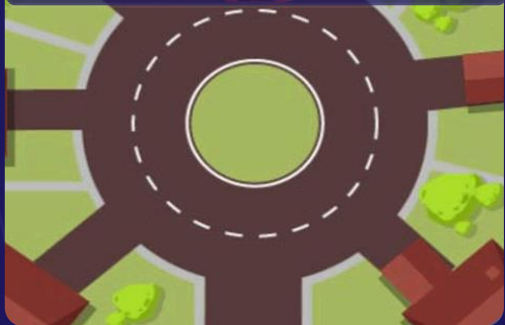
Change in **state of motion**



Change in **speed**



Change in **direction**



Change in **shape**



1.3 Types of Force

►►► Contact Force

Frictional force



- Resists motion when the two or more surfaces come in contact.

Normal force



- The force that a surface employs on any other body.

Muscular force



- When any force applied by using of muscles like arms or legs.

►►► Non-Contact Force

Gravitational force



- The force of attraction between all masses in the universe.

Electrostatic force



- An attractive and repulsive force between particles are caused due to their electric charges.

Magnetic force



- An attractive or repulsive force that is exerted between the poles of a magnet.

1.4 Balanced and Unbalanced Force

Net force is the **equivalent** force acting on the object.



$$F_{\text{net}} = F_1 + F_2$$



$$F_{\text{net}} = F_1 - F_2$$

▶▶▶ Balanced force

$$F_{\text{net}} = 0$$



$$F_{\text{net}} = F_1 - F_2 = 0$$

- All forces **cancel** out each other.
- Net force is **zero**.

▶▶▶ Unbalanced force

$$F_{\text{net}} \neq 0$$



$$F_{\text{net}} = F_1 - F_2 \neq 0$$

- Forces **do not** cancel out.
- Net force is **not equal to zero**.

2. Newton's Laws of Motion

2.1 First Law of Motion

- An object remains in a state of **rest** or in a state of **uniform motion**, until and unless an **external force** acts on it



Rest : No change in **position**.

Uniform motion : No change in **velocity**.

2.2 Inertia



- Resistance** to the change in **state of rest** or **uniform motion**.
- The **mass** of an object is a measure of its **inertia**.

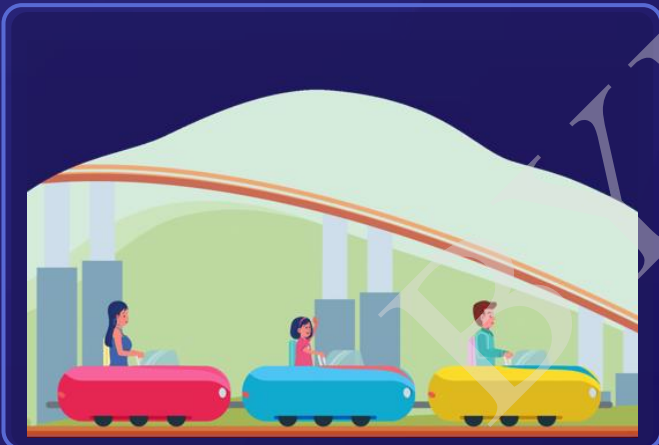
2.3 Types of Inertia

►►► Inertia of rest



- **Resistance** offered by the body to continue in the state of **rest**.

►►► Inertia of motion



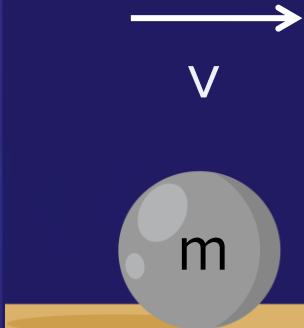
- **Resistance** offered by the body to continue in the state of **motion**.

►►► Inertia of direction



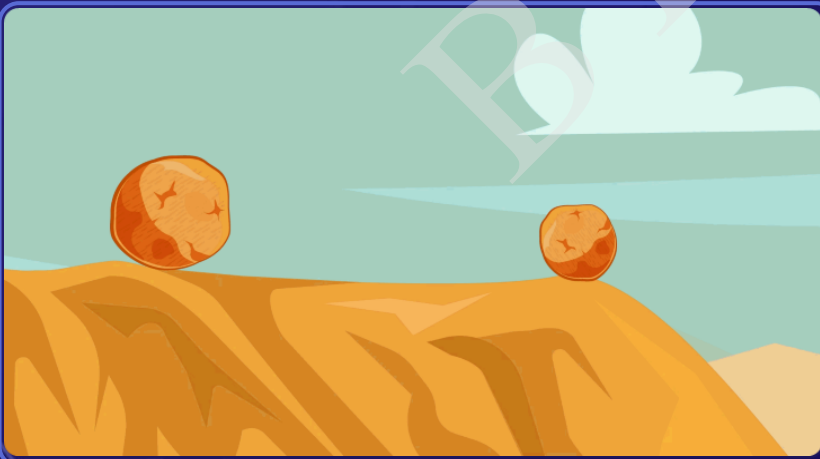
- **Resistance** offered by the body to continue moving in **same direction**.

2.4 Momentum



- Product of **Mass** and **Velocity**
- Momentum (p) = Mass (m) x Velocity (v)
- S.I. unit = **kg m/s** or **N s**
- **Vector** quantity

2.5 Second Law of Motion



- **Net force** acting on a body equals the rate of **change** of **momentum**.

$$F_{\text{net}} = \frac{\Delta p}{\Delta t}$$

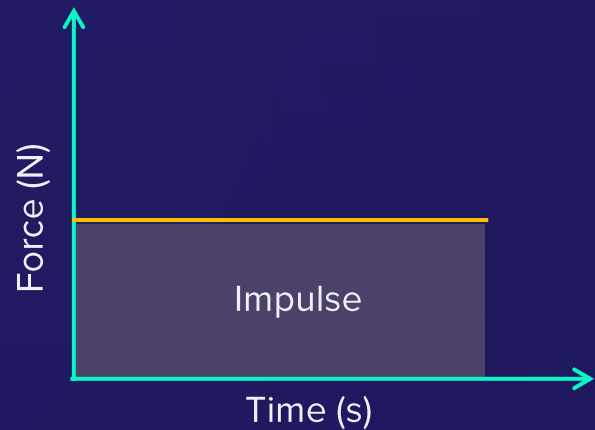
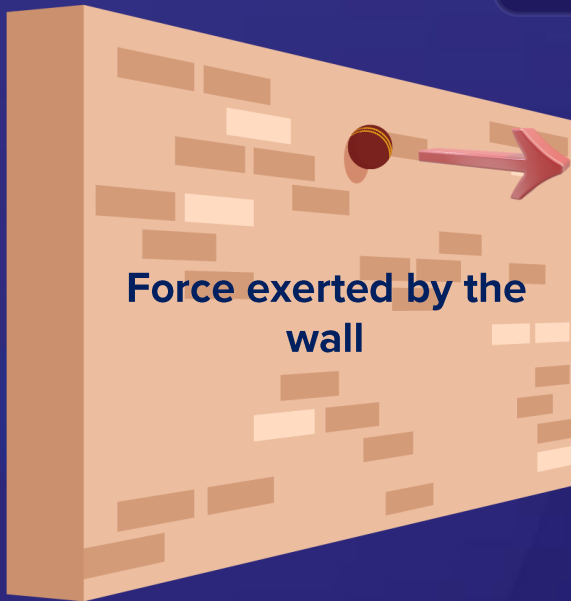


For constant mass

$$F_{\text{net}} = \frac{m \Delta v}{\Delta t}$$

$$F_{\text{net}} = ma$$

2.6 Impulse

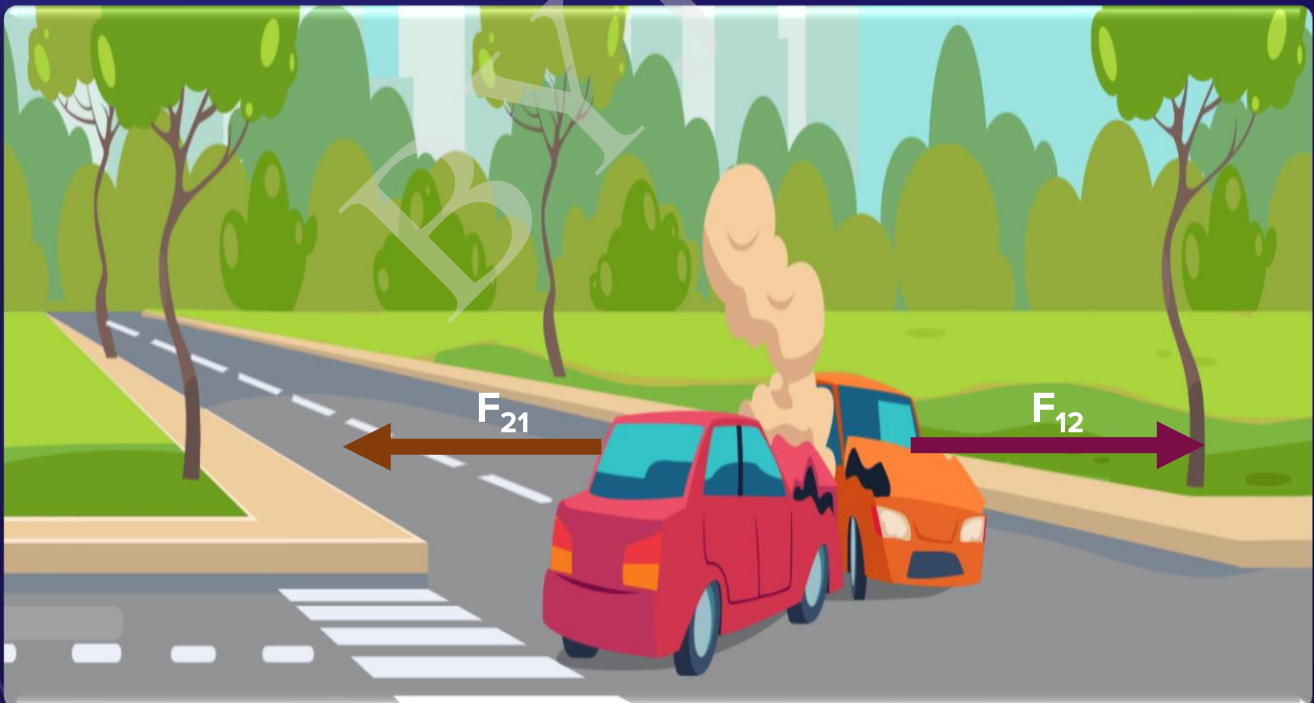


Impulse = Change in momentum

$$J = mv - mu = m \Delta v$$

$$J = F \times t$$

2.7 Third Law of Motion



- For every **action** there is an equal and opposite **reaction**.
- The action and reaction force always act on **different bodies**.

$$F_{12} = - F_{21}$$

Applications of 3rd Law

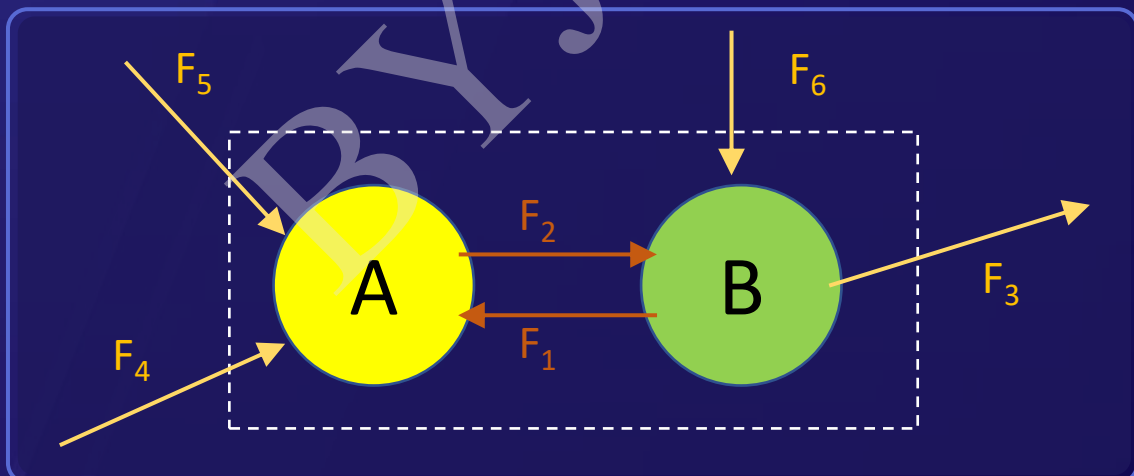


Recoil of a gun



Propulsion of a rocket

2.8 Internal and External Forces



Internal forces: F_1, F_2
External forces: F_3, F_4, F_5, F_6

- Only external forces are capable of producing motion in a body.

Formula Sheet

1

MOMENTUM

$$p = mv$$

2

SECOND LAW
OF MOTION

$$F_{\text{net}} = \frac{\Delta p}{\Delta t}$$

$$F_{\text{net}} = ma$$

3

IMPULSE

$$J = mv - mu = m \Delta v$$

$$J = F \times t$$

4

THIRD LAW
OF MOTION

$$F_1 = -F_2$$

5

CONSERVATION OF
MOMENTUM

$$m_1 u_1 + m_2 u_2 = m_1 v_1 + m_2 v_2$$

